

MAPPING CLASS GROUPS

These are problems about mapping class groups, with a focus on braid groups. See also problems [\(O10\)](#), [\(E4\)](#).

*(B1) Are braid groups linear? [Background](#)

(B2) Is the Gassner representation of the pure braid group P_n faithful for every n ? [Background](#)

(B3) Is the Burau representation of the braid group B_n faithful for $n = 4$? [Background](#)

(B4) (J.Birman) Give necessary and sufficient conditions for a square matrix over Laurent polynomial ring to be the Burau matrix of some braid.

(B5) (V.Lin) Let $n > 4$. (a) Does the braid group B_n have a non-trivial non-injective endomorphism?

* (b) Is it true that every non-trivial endomorphism of the commutator subgroup $[B_n, B_n]$ is actually an automorphism? [Background](#)

* (B6) (V.Lin) Let $n > 4$. (a) Does the braid group B_n have a proper torsion-free non-abelian factor group?

(b) Does the commutator subgroup $[B_n, B_n]$ have a proper torsion-free factor group? [Background](#)

* (B7) (V.Lin) Let $n > 4$. (a) Is it true that every automorphism of the commutator subgroup $[B_n, B_n]$ can be extended to an automorphism of the whole group B_n ?

(b) Is it true that every non-trivial endomorphism of $[B_n, B_n]$ can be extended to an endomorphism of B_n ? [Background](#)

(B8) (P. Dehornoy) We call a braid word w σ -positive (resp. σ -negative) if the generator σ_i with minimal index occurs only with positive (resp. negative) exponents in w . Is it true that, for every n , there exists a constant $c(n)$ such that every n -strand braid word of length N is equivalent to a σ -positive or a σ -negative braid word of length $c(n)N$ at most?

(B9) (P. Dehornoy) Say that a braid is special if it can be obtained from the trivial braid by using iteratively the self-distributive exponentiation $a \frown b = a S(b) \sigma_1 S(a)^{-1}$, where S is the shift endomorphism that maps σ_i to σ_{i+1} for every i . How many special braids are there in B_n ?

*(B10) (G.Makanin) Is it true that in any group B_n , $g^k = h^k$ for some $k \neq 0$ implies that g is conjugate to h ? [Background](#)

(B11) (V.Shpilrain) Let K_n be the kernel of the Burau representation of the braid group B_n , $n > 4$. Is the factor group B_n/K_n torsion free? [Background](#)

(B12) Do braid groups B_n , $n > 4$, have non-elementary hyperbolic factor groups? [Background](#)

(B13) Is there an algorithm with subquadratic worst-case time complexity that would solve the word problem in braid groups? [Background](#)